

Week 8, Lecture 1: Priors and Posteriors

Instructor notes. Part 1: prep reading. Part 2: podium cheat sheet

How to prepare (for students)

- **Big picture.** On Wednesday we built Bayes' rule for parameter values and worked out why a Beta prior and a binomial likelihood give a Beta posterior. Today we use that result on a real example with two different priors, learn how to summarize a posterior, and talk about common misunderstandings and why Bayesian methods are so widely used. If there's time, we'll start on optimization: how `optim()` finds the best parameter values.
- **Read:** Bolker 6.2.2 (printed p. 237-243) and 6.4.2 (p. 258-260), if you haven't yet. To get ahead on optimization: Bolker 7.1-7.2.
- **Review:** your notes on Beta-binomial conjugacy from Wednesday, the Beta distribution (Bolker 4.5.2.5, p. 176), and the log-odds, $\log(p/(1-p))$, from Wednesday's logistic regression opening.

Overview (instructor only)

This file has two parts, both instructor-only. **Part 1** is read beforehand. **Part 2** (after the page break, large print) is the podium copy.

Beat	Time
WOD (to be planned with the rest of Week 8)	~8 min
Lecture	~32 min
Discussion	~8 min

Where this came from. Sections 2-5 and the discussion were planned for Week 7 Wednesday, which instead covered logistic regression, Bayes' rule for parameters, and the conjugacy math (see `NOTES_Week7_Lecture2_BayesianInference.Rmd`). Start with the germination example.

Pacing. Both Week 7 lectures covered about half of what was planned. Core: sections 1-2, confusions (a), (b), (e) from section 4, two reasons from section 5, and section 7. Section 3 and the rest of 4-5 are "if time"; section 6 (optimization) only if the Bayesian material finishes early, otherwise it opens Wednesday.

WOD. Not yet chosen. Candidate: a shortened version of `wod/WOD_Week7_Lecture2_CurvatureByHand.Rmd` which was never handed out in Week 7 (rename it to its new week/slot if used).

Source material: Bolker 4.3.3 (priors, and why flat priors aren't neutral), 6.2.2.1 (binomial/Beta conjugate example; the numbers below are his, recast as a germination trial), 6.4.2 (credible intervals), 7.1-7.2 (optimization); legacy `LECTURE6.Rmd` ("Bayesian Analysis #1: Concepts": mean vs. mode, credible intervals).

Example choice. The conjugate example is a seed-germination trial rather than Bolker's tadpoles. The numbers (30 of 40, and a prior worth 120 of 200) are Bolker's from p. 237-238, so students who did the reading can check them, but the setting is one the agricultural and restoration students will recognize.

Where this sits in the week

- **Last week:** likelihood intervals (Monday), the quadratic approximation and profiles (Lab 6), then logistic regression and the start of Bayesian inference (Wednesday).
- **Today:** finish the one-parameter Bayesian material; start optimization if time.
- **Tuesday:** group meetings on final project proposals (no lab).
- **Wednesday:** optimization (Bolker Ch. 7).
- **Coming up:** MCMC and Lab 7 on Bayesian inference (Week 9).

Part 1: Lecture content

1. Big picture and recap (~3 min)

Loop on the board:

THEORY $\rightarrow$ MODEL(S) $\rightarrow$ DATA (fit) $\rightarrow$ COMPARE $\rightarrow$ repeat

- Same model, same likelihood; Bayesian inference describes our uncertainty about a parameter with a probability distribution.
- Recap: posterior $\propto$ likelihood $\times$ prior. With a Beta prior and binomial data, the posterior is $\text{Beta}(\alpha + k, \beta + n - k)$.
- Today's question: how much does the prior matter, and what do we report?

2. A worked example: germination with two priors (~10 min)

Setting. A restoration crew tests a native seed lot: 30 of 40 seeds germinate. What is the lot's germination probability p ?

The likelihood is binomial: $\mathcal{L}(p) \propto p^{30}(1-p)^{10}$.

Recap from Wednesday. With a $\text{Beta}(\alpha, \beta)$ prior, the posterior is $\text{Beta}(\alpha + k, \beta + n - k)$: just add counts.

- Interpretation (Bolker p. 237): $\alpha - 1$ and $\beta - 1$ act like earlier successes and failures. Beta(1, 1) is flat: no prior seeds.
- **Flat prior:** posterior Beta(31, 11).
- **Informative prior:** the supplier's records say 120 of 200 seeds from earlier lots germinated. Prior Beta(121, 81); posterior Beta(151, 91).

```

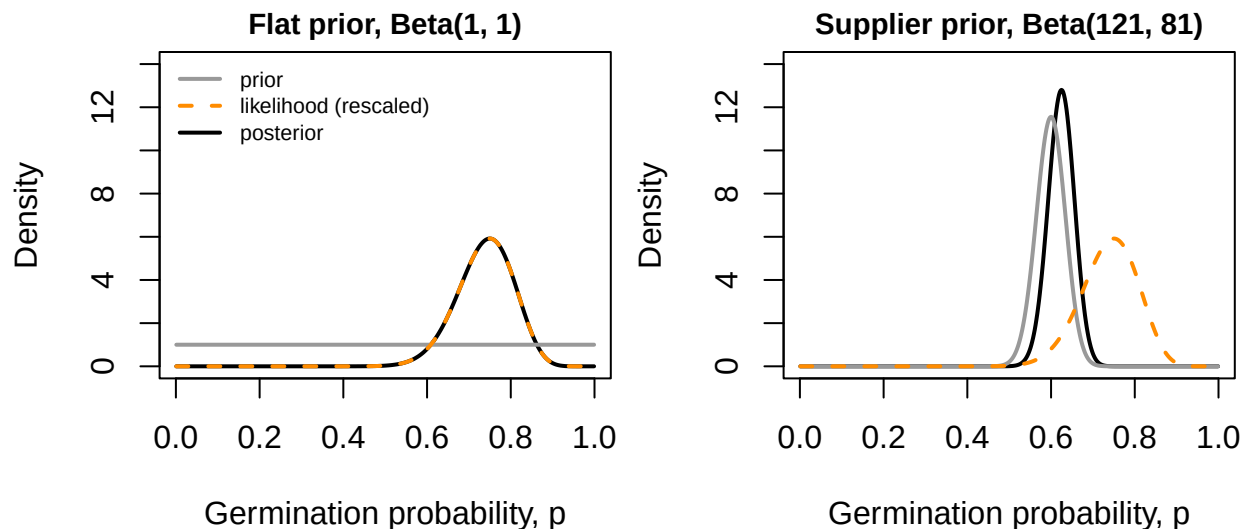
k <- 30; n <- 40
post_flat <- c(1 + k, 1 + n - k)
post_info <- c(121 + k, 81 + n - k)
beta_mean <- function(ab) ab[1] / sum(ab)
beta_mode <- function(ab) (ab[1] - 1) / (sum(ab) - 2)
summ <- rbind(
  flat = c(mean = beta_mean(post_flat), mode = beta_mode(post_flat),
    qbeta(c(0.025, 0.975), post_flat[1], post_flat[2])),
  informative = c(mean = beta_mean(post_info), mode = beta_mode(post_info),
    qbeta(c(0.025, 0.975), post_info[1], post_info[2]))
)
colnames(summ)[3:4] <- c("lower95", "upper95")
round(summ, 3)

##           mean  mode lower95 upper95
## flat       0.738 0.750   0.597   0.858
## informative 0.624 0.625   0.562   0.684

f2 <- function(v) formatC(v, format = "f", digits = 2)

p_seq <- seq(0.001, 0.999, length.out = 500)
lik_scaled <- dbeta(p_seq, k + 1, n - k + 1) # likelihood rescaled to area 1
plot_panel <- function(prior_ab, post_ab, title) {
  plot(p_seq, dbeta(p_seq, post_ab[1], post_ab[2]), type = "l", lwd = 2,
    ylim = c(0, 14), xlab = "Germination probability, p", ylab = "Density",
    main = title, cex.main = 0.9)
  lines(p_seq, dbeta(p_seq, prior_ab[1], prior_ab[2]), lwd = 2, col = "grey60")
  lines(p_seq, lik_scaled, lwd = 2, lty = 2, col = "darkorange")
}
par(mfrow = c(1, 2), mar = c(4, 4, 1.5, 0.5))
plot_panel(c(1, 1), post_flat, "Flat prior, Beta(1, 1)")
legend("topleft", c("prior", "likelihood (rescaled)", "posterior"),
  lwd = 2, lty = c(1, 2, 1), col = c("grey60", "darkorange", "black"),
  bty = "n", cex = 0.7)
plot_panel(c(121, 81), post_info, "Supplier prior, Beta(121, 81)")

```



- **Flat prior:** the posterior *is* the rescaled likelihood. Its mode is 0.75, the MLE; its mean is 0.738, slightly pulled toward 0.5.
- **Supplier prior:** the posterior sits between prior and data, closer to the prior (mean 0.624), because the prior is “worth” 200 seeds and the data only 40.
- **More data wins.** If the crew had tested 400 seeds with 300 germinating, the same supplier prior gives a posterior mean of 0.699: the data now outweigh the prior two to one.
- **Updating in sequence.** Today’s posterior is tomorrow’s prior. Testing 40 seeds and then 40 more gives the same answer as testing 80 at once (Bolker p. 237).

3. Summarizing a posterior (~4 min)

- **Point estimates.** Posterior mean (most common), median, or mode. With a flat prior, the mode equals the MLE. For skewed posteriors the three differ; Bolker prefers the mean because it uses the whole distribution.
- **Credible interval.** The central 95% of the posterior (2.5% and 97.5% quantiles), or the highest posterior density (HPD) interval, the shortest interval holding 95% (Bolker p. 258-259). For symmetric posteriors they coincide.
- **Interpretation.** “Given the model, the prior, and the data, there is a 95% probability that p is between 0.60 and 0.86.” That is the sentence most people *want* to say about a confidence interval (Week 7 Monday’s discussion question 1), and a Bayesian is allowed to say it.

4. Where new Bayesians get stuck (~8 min)

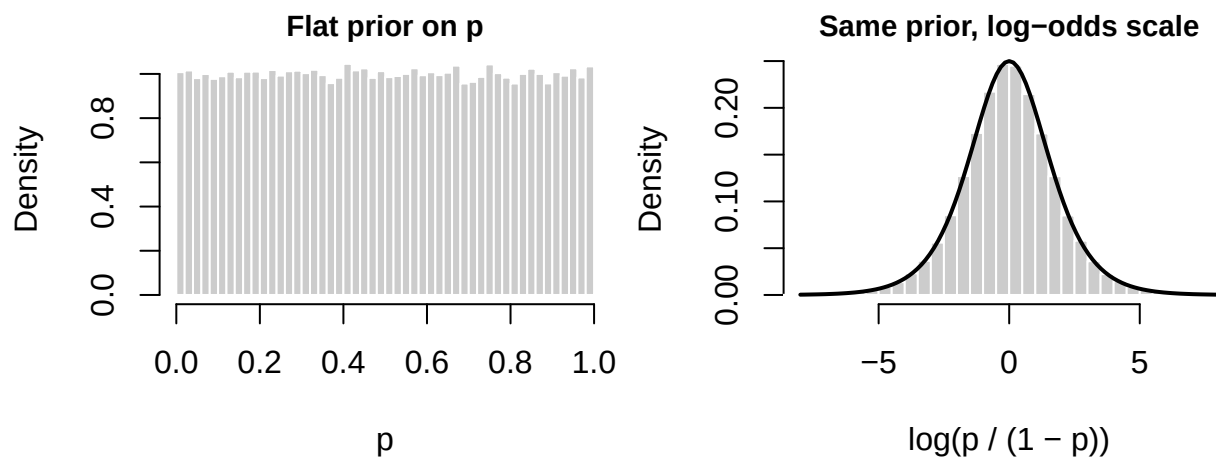
Core: (a), (b), (e). The rest if time; each takes a minute or two. Callback for (b): the log-odds from Wednesday’s logistic regression opening.

(a) “**The likelihood is the posterior.**” Only after multiplying by a prior and rescaling to area 1. The likelihood alone doesn’t integrate to 1 (Week 5 WOD) and isn’t a distribution

for p .

(b) **“A flat prior means no information.”** Flat on one scale isn’t flat on another (Bolker 4.3.3, the bird-mass example). A uniform prior on p implies a prior on the log-odds, $\log \frac{p}{1-p}$, that is peaked at 0:

```
set.seed(746)
p_draws <- runif(1e5)
par(mfrow = c(1, 2), mar = c(4, 4, 1.5, 0.5))
hist(p_draws, breaks = 40, freq = FALSE, col = "grey80", border = "white",
     main = "Flat prior on p", xlab = "p", cex.main = 0.9)
hist(qlogis(p_draws), breaks = 60, freq = FALSE, col = "grey80", border = "white",
     main = "Same prior, log-odds scale", xlab = "log(p / (1 - p))",
     xlim = c(-8, 8), cex.main = 0.9)
curve(dlogis(x), add = TRUE, lwd = 2)
```



So “uninformative” always means “uninformative on some scale.” With plenty of data it rarely matters; with little data it can.

(c) **“Priors let you get any answer you want.”** In principle yes (Bolker p. 150: a prior of 0 on a hypothesis means no data can ever support it). In practice: priors must be reported, reviewers can object, and the standard move is a **prior sensitivity analysis**: rerun with a few reasonable priors and see whether the conclusion changes. If it does, the data are weak and that’s worth knowing.

(d) **“Bayesian means MCMC.”** MCMC is a computing method for approximating posteriors when the denominator can’t be computed. The germination example needed none. Conversely, MCMC can be used for non-Bayesian problems too.

(e) **“The credible interval and the confidence interval are the same thing.”** With a flat prior and plenty of data the numbers are often close (Week 7 pikas: likelihood ratio [1.82, 4.61]; with a flat prior the posterior is $\text{Gamma}(19, 6)$, credible interval [1.91, 4.74]), but the statements differ: a credible interval is a probability statement about the parameter; a confidence interval is a statement about the procedure.

(f) “**A high posterior probability proves the model.**” Every posterior is conditional on the model being right. A Bayesian model still has to be confronted with data (does it predict data that look like what we saw?) and compared with rivals. The loop on the board doesn’t change.

5. Why Bayesian methods are so widely used (~5 min)

Interpretation.

- Direct probability statements about parameters and hypotheses: “95% probability the germination rate is between 0.60 and 0.86.”
- Natural accumulation of evidence: last year’s posterior is this year’s prior.

Practical.

- **Uncertainty in derived quantities is easy to get.** Callback to Lab 6: for the LD50, a Bayesian draws many (a, b) pairs from the posterior and computes $-a/b$ for each; the spread of those values is the posterior for the LD50. No profile, no reparameterization.
- **Hierarchical models.** Plots within sites within regions, animals measured repeatedly, many species sharing information: these models have many parameters, some describing variation among groups. They are awkward to fit by maximum likelihood and natural in a Bayesian framework (later in the semester).
- **Hidden quantities.** True occupancy when detection is imperfect (the frog from Lab 5), true infection status behind a noisy test, the true population size behind a count. These are just more unknowns in a Bayesian model.
- **Small samples.** No reliance on large-sample approximations like the Hessian interval, which goes negative for the Week 7 chytrid data (2 of 20 positive).
- **Prior information.** Previous studies, expert knowledge, or physical limits can be built in openly.

Costs. Priors must be chosen and defended; computation can be slow; MCMC output has to be checked for convergence.

6. Optimization preview (~5 min, only if time)

Bolker 7.1-7.2. Wednesday develops this; today is just the motivation.

- Every MLE so far came from calculus (one parameter), a grid, or `optim()` used as a black box.
- Grids don’t scale: 100 values per parameter is 100 evaluations for one parameter, $100^{10} = 10^{20}$ for ten.
- Optimizers search instead: start somewhere, use the slope (and maybe the curvature, the Hessian from Lab 6) to decide where to step next, and stop when no step improves the fit.
- Teaser for Wednesday: why `optim()` sometimes stops in the wrong place, and why starting values matter.

7. Return to the big picture (~1 min)

- Same model, same likelihood. Bayesian inference multiplies by a prior and reads uncertainty off a posterior distribution.
- With one parameter and a conjugate prior, it's just adding counts. With many parameters, we need MCMC (Week 9).

Discussion (~8 min)

Pick one or two:

1. With the supplier prior, the posterior mean germination rate is 0.62; with a flat prior it's 0.74. A restoration manager says, "Keep the supplier's numbers out of my estimate." Is that reasonable? What would you report to them?
2. Two analysts fit the same germination data with "flat" priors, one flat on p and one flat on the log-odds of p . Could they get different answers? When would it matter?
3. In Lab 6 you built a profile interval for the LD50. How would a Bayesian get an interval for the LD50, and which approach would you rather use for a model with ten parameters?
4. The Week 7 likelihood ratio interval (95%) for the pikas was about [1.8, 4.6]. A land manager asks, "So there's a 95% chance the true density is between 1.8 and 4.6?" Can you say yes? What could a Bayesian say instead?

Question 1 surfaces the "priors are cheating" worry. The manager has a point (the prior is worth 200 seeds to the data's 40, and it comes from other lots and an interested party), but the earlier lots aren't worthless either. Where to land: report both posteriors (a prior sensitivity analysis), perhaps with a down-weighted prior in between (the supplier's 60% scaled down to 12 of 20 seeds, Beta(13, 9)). Whether past lots are relevant to this one is an ecological question, not a statistical one. Question 3 connects today to Lab 6 and sets up MCMC.

PART 2: PODIUM CHEAT SHEET

Terse on purpose. Full rationale and code: Part 1.

WOD ~8 min (TBD). Lecture ~32 min. Discussion ~8 min.

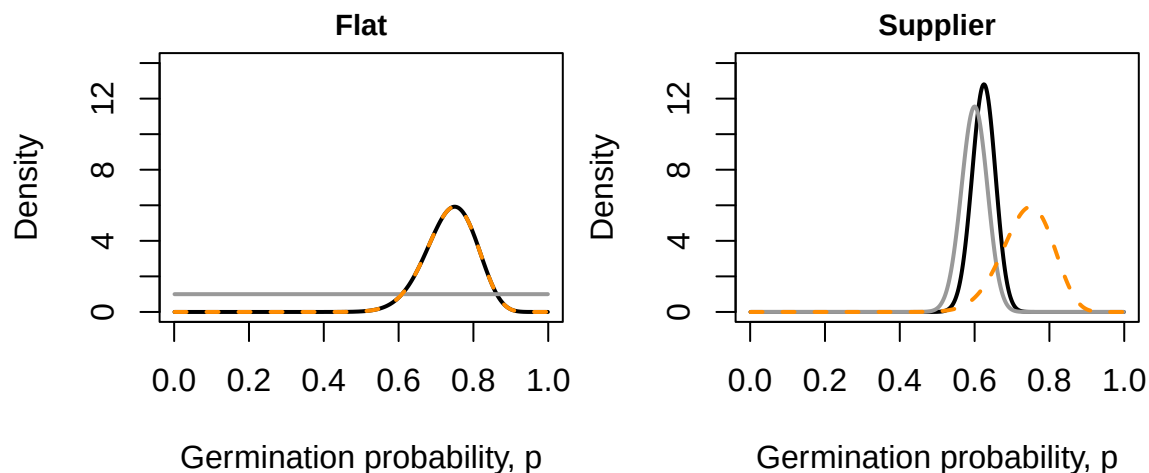
1. Big picture and recap (~3 min)

- Loop on board: THEORY $\rightarrow$ MODEL $\rightarrow$ DATA (fit) $\rightarrow$ COMPARE $\rightarrow$ repeat
- Posterior $\propto$ likelihood $\times$ prior
- Beta(α, β) + k of $n \rightarrow$ Beta($\alpha + k, \beta + n - k$)
- Today: how much does the prior matter?

2. Germination, two priors (~10 min)

Seeds: **30 of 40** germinate. Supplier: 120 of 200.

- Beta prior $\times$ binomial $\rightarrow$ Beta($\alpha + k, \beta + n - k$)
- Conjugate = add counts; $\alpha - 1, \beta - 1$ = prior successes, failures
- Flat: Beta(31, 11); mean 0.738, mode 0.75 = MLE
- Supplier: Beta(151, 91); mean 0.624
- 300 of 400 with supplier prior: mean 0.699
- Sequential updating = same answer

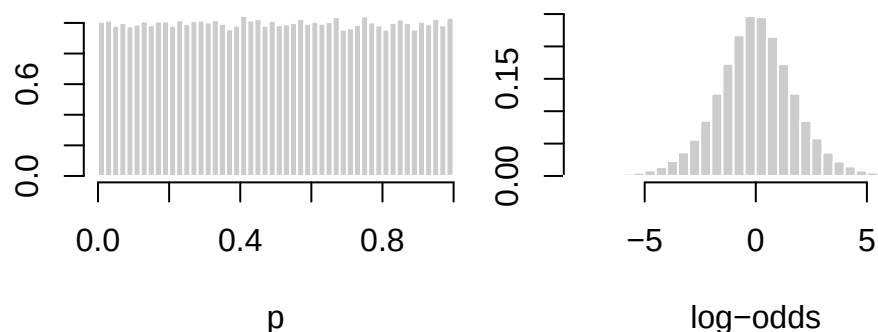


3. Summaries (~4 min)

- Mean (usual), median, mode (= MLE if flat)
- Credible interval: quantiles or HPD
- Flat: [0.60, 0.86]; supplier: [0.56, 0.68]
- “95% probability p is in here” is allowed

4. Where people get stuck (~8 min)

- (a) Likelihood $\neq$ posterior (Week 5 WOD)
- (b) Flat isn't flat on every scale



- (c) Priors: report them; sensitivity analysis
- (d) Bayes $\neq$ MCMC
- (e) Credible $\neq$ confidence (pikas: [1.82, 4.61] vs. [1.91, 4.74])
- (f) Posterior is conditional on the model; still check and compare

5. Why so popular (~5 min)

- Direct probability statements
- Evidence accumulates: posterior $\rightarrow$ next prior
- Derived quantities: LD50 = $-a/b$ for each posterior draw (Lab 6)
- Hierarchical models (later)
- Hidden quantities: occupancy, true infection status
- Small samples: no Hessian approximation (chytrid)
- Costs: priors, computation, convergence

6. Optimization preview (only if time)

- MLEs so far: calculus, grids, `optim()` as a black box
- Grids: 100 values/parameter $\rightarrow 10^{20}$ evaluations for 10 parameters
- Optimizers: start, use slope (and curvature) to step, stop
- Wednesday: why `optim()` can stop in the wrong place

7. Wrap (~1 min)

- Conjugate = add counts; many parameters = MCMC (Week 9)

Discussion (~8 min)

1. Supplier prior: mean 0.62; flat: 0.74. Manager: “keep the supplier’s numbers out.” Reasonable? What to report?
2. Flat on p vs. flat on log-odds: different answers?
3. Bayesian interval for the LD50? Which approach for 10 parameters?
4. Week 7 pikas [1.8, 4.6]: “95% chance”? Can you say yes?